

ECONOMIC STRUCTURES FOR LOW-DEMAND FUTURES: A SYSTEMATIC MAPPING

ESEE 2026

Proposed for Best Student Paper

Kilian Rouge & Gabriele Dabbaghian

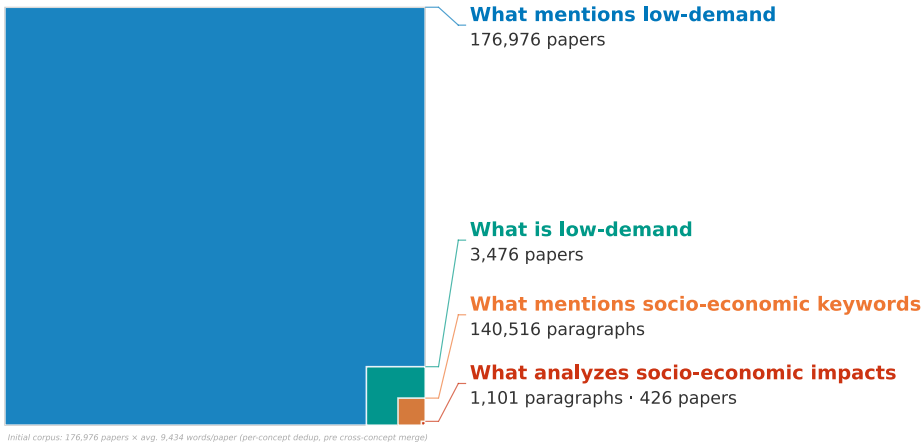
PhD Candidates

CIREN, Paris

July 2026



IN A NUTSHELL



► Visual representation of corpus size at each step (area is proportional to word count)

1. INTRODUCTION

LOW-DEMAND MITIGATION IS EXPANDING

- Low-demand mitigation is an essential complement to efficiency and consistency but it's socio-economic implications remain poorly documented



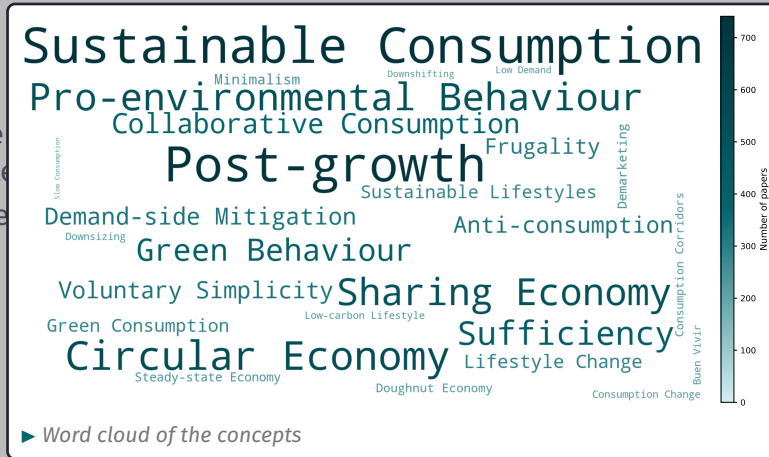
LOW-DEMAND MITIGATION IS EXPANDING

- Low-demand mitigation is an essential complement to efficiency and consistency but it's socio-economic implications remain poorly documented
- This idea can be found across different **concepts**



LOW-DEMAND MITIGATION IS EXPANDING

- Low-demand mitigation is expanding
- This idea is documented



LOW-DEMAND MITIGATION IS EXPANDING

- Low-demand mitigation is an essential complement to efficiency and consistency but it's socio-economic implications remain poorly documented
- This idea can be found across different **concepts**
- And it includes a wide range of **practices**



LOW-DEMAND MITIGATION IS EXPANDING

- Low-demand mitigation is consistently the most documented
- This idea is expanding
- And it is becoming more specific

Absolute Reduction

Supply Side

Shift

Usage Behaviour

Buying Patterns

Curtailment

Non-specific Behaviour Change

Number of papers

► Word cloud of the practices (Sandberg, 2021; Schanes et al., 2016)

LOW-DEMAND TRANSITIONS AS DIRECTED STRUCTURAL CHANGE

To better understand their socio-economic implications we propose to analyze low-demand transitions as a process of directed structural change constrained by physical limits



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Evolution in the way the productive activity is organized

- Sectoral composition
- Industrial organization
- Demand
- Employment dynamics
- Technical change
- Institutions

1. Low-demand affects **several economic structures in interaction**
2. Micro: Understanding adoption mechanisms doesn't say if the transition closes into a **coherent configuration**
3. Aggregate: suppresses **compositional and distributional information**



LOW-DEMAND TRANSITIONS AS DIRECTED STRUCTURAL CHANGE

To better understand their socio-economic implications we propose to analyze low-demand transitions as a process of **directed structural change** constrained by physical limits

Deliberate reduction in final demand as driving force of the transition

Evolution in the way the productive activity is organized

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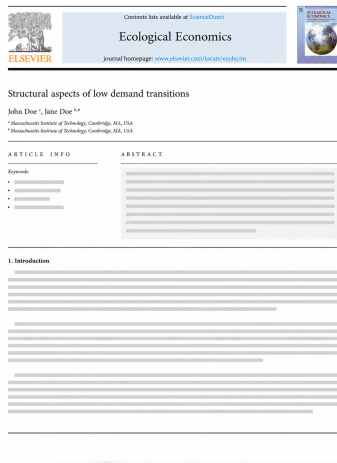
2. METHODOLOGY

SCREENING METHODOLOGY

1. **Identifying low-demand concepts**
27 concepts
2. Search on Scopus & WoS
176,976 papers
3. Screening abstracts on low-demand understanding
3,476 included
4. Searching 373 keywords related to structural dimensions in full text
140,516 paragraphs
5. Screening & extracting paragraphs
1,101 included

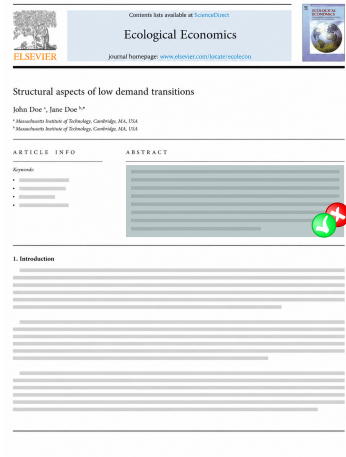


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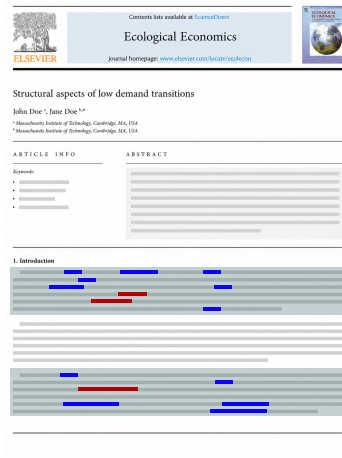
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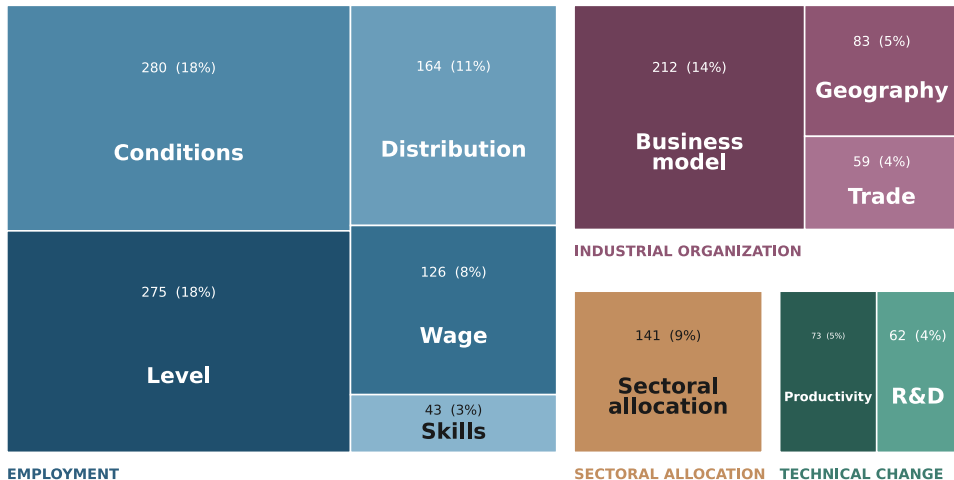
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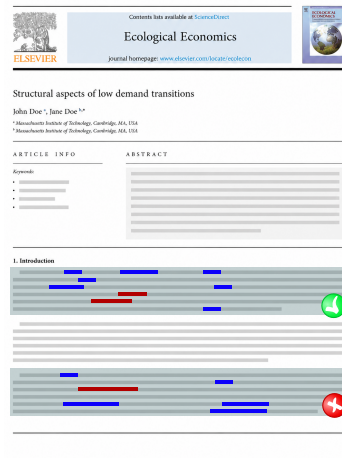
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► Structural dimensions analyzed (Ciarli & Savona, 2019) and related count of occurrences

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3. RESULTS

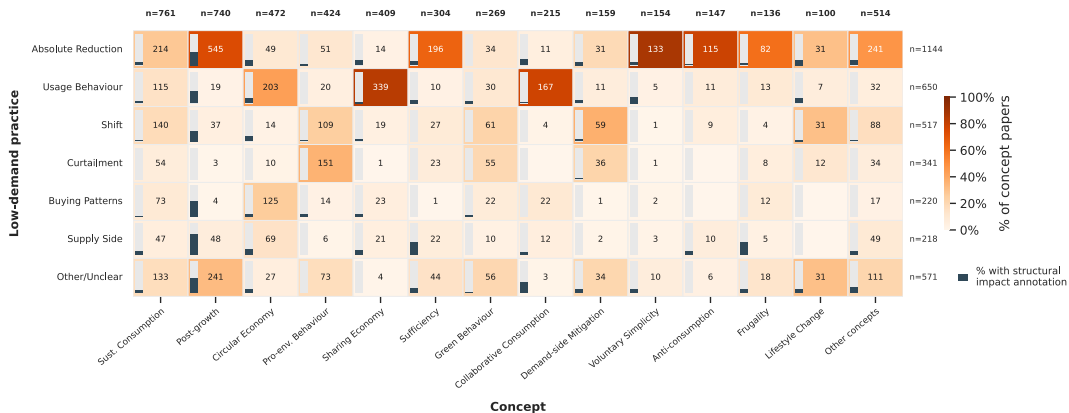
KEY RESULTS

- 1. A fragmented literature, siloed by concept and practice.**
2. Structural impacts remain a minority concern: only 12% of papers
Most papers discuss demand reduction without engaging macro-level effects.
3. Dimensions studied in isolation, cross-cutting analysis is rare
Sectoral and technical change are rarely analyzed in isolation.
4. A largely normative literature, few empirics and few models
Opinions or references to a small set of foundational mechanism papers.
5. Convergence on a handful of analytical mechanisms
Productivity trap, profit-squeeze, Baumol cost disease, I/O linkages, working-time absorption.
6. Convergence on a policy menu
Progressive taxes, basic income, universal basic services, working-time reduction, public ownership.



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► Distribution of low-demand papers and structural impacts across concepts and practices

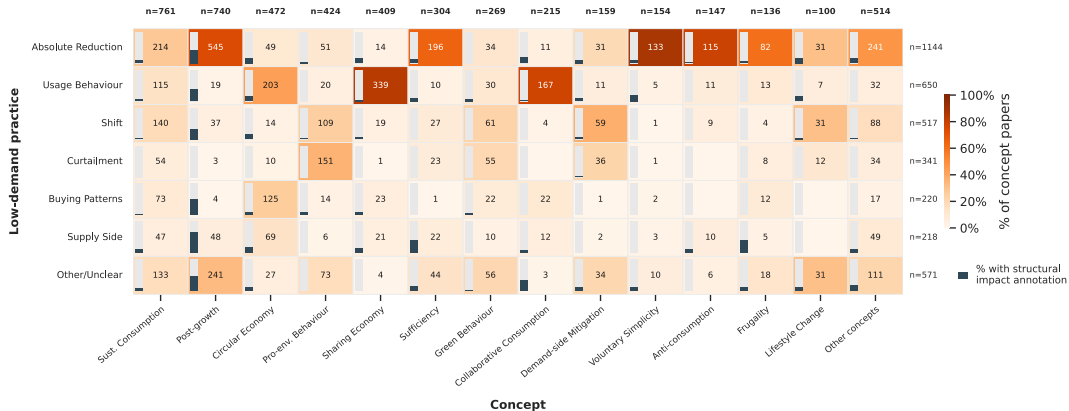
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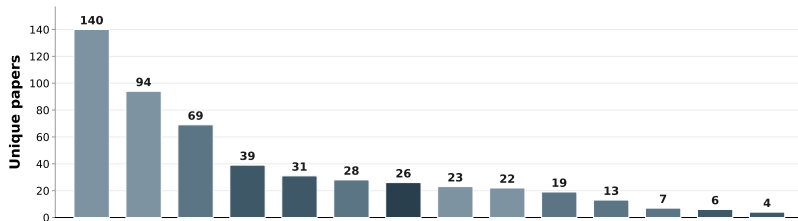
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KEY RESULTS



Employment

Conditions, Level, Distribution, Wage, Skills

Industrial organization

Business model, Geography, Trade

Sectoral allocation

Technical change

R&D, Productivity

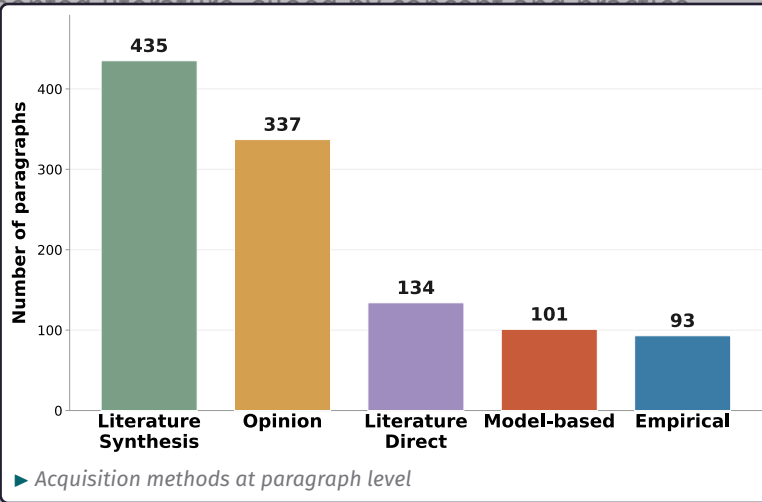
► Co-occurrence of structural dimensions across papers

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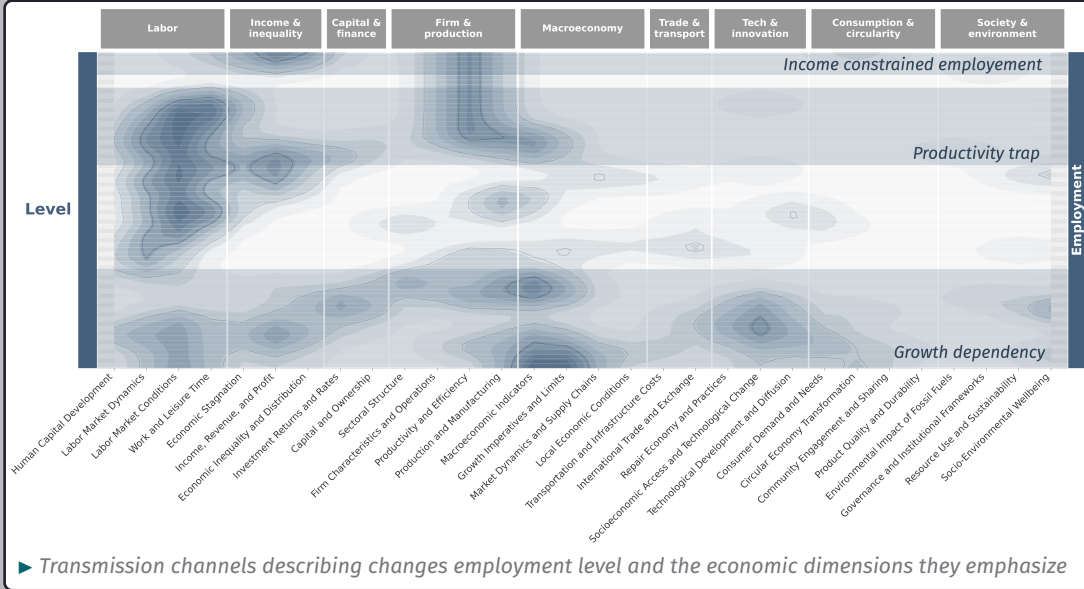


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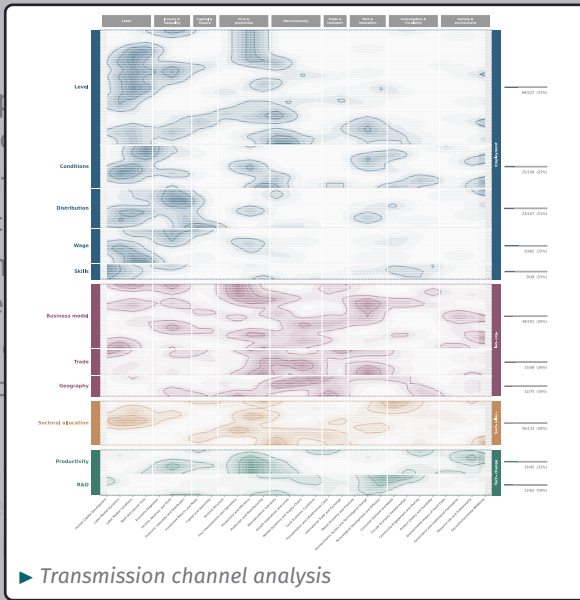


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4. CONCLUSION

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Main contributions

1. Cover a **large set of low-demand concepts and practices** thanks to advances in large systematic review methods
2. Propose **directed structural change as a framework** to better understand low-demand transitions and their socio-economic implications



APPENDIX

BIBLIOGRAPHY (I)

- Ciarli, T., & Savona, M. (2019). Modelling the Evolution of Economic Structure and Climate Change: A Review. *Ecological Economics*, 158, 51–64.
- Sandberg, M. (2021). Sufficiency transitions: A review of consumption changes for environmental sustainability. *Journal of Cleaner Production*, 293, 126097.
- Schanes, K., ... Hertwich, E. (2016). Low carbon lifestyles: A framework to structure consumption strategies and options to reduce carbon footprints. *Journal of Cleaner Production*, 139, 1033–1043.

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PhD Candidates

Ghent, July 2026